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Commentary

Improvingresearchtransparency:Aninterpretationoftheupdated consolidatedstandardsofreportingtrials202

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Introduction

Well-designed,strictlyimplemented,andfullystandardizedrandomizedcontrolledtrials(RCTs)areaprerequisitefordevelopingreliablescientificevidence,whichcanimproveclinicalpractice,healthoutcomes,andultimatelybenefitpatients.Suboptimalreportingispervasiveinmedicalresearch,resultinginbiasedresearchrecordsandpersistentuncertaintyaboutthequalityofavailableevidence.[1–4] Thestandardizationofresearchreportshasattractedconsiderable attention.In1996,theConsolidatedStandardsofReportingTrials (CONSORT)wasfirstpublishedtoimprovethequalityofRCTsand enhancethereproducibilityoftrialmethods,results,andinferences.[5] CONSORTprovidesimportantguidelinesforreviewers,editors,and readerstoassessthereliabilityandauthenticityofstudyfindingsand hasbeenwidelypraisedinpracticalapplications.[6] Sincethen,CONSORThasbeenrevisedthreetimes:in2001,[7] 2010,[8] andthecurrent updatein2025.[4] TheCONSORTgrouphasworkedtirelesslytopromote standardizedandtransparentclinicaltrialreporting.[9] WiththeCONSORTstatement,clinicaltrialreportinghasimprovedsignificantly, featuringstructuredandtransparentdocumentationinsteadofarbitrary inference,balancedemphasisonbothprocessandoutcomes(ratherthan beingresult-oriented),andintegrationofisolatedstudiesintocredible scientificevidence.Duringthesechanges,CONSORTmadesignificant contributionsandplayedaleadingandexemplaryroleinthe

standardizationofclinicalresearchreportsforotherclinicalstudy designs.

Thedevelopmentofantitumordrugsiscloselyintertwinedwithhigh-qualityclinicaltrials,makingitaglobalresearchfocus.Withtherapid developmentofrelatedbasicandclinicalresearchrecently,manyclinical oncologystudieshavebeenconducted,andremarkableadvanceshave beenmadeintheresearchanddevelopmentofantitumordrugsin China.[10]Asof2020,approximatelyone-thirdofglobalclinicaltrialshave beenconductedinChina,andthisproportionhasincreasedsignificantly, especiallyforoncologyclinicaltrials.[11,12]However,oncologytrialsare characterizedbyuniquechallengesintheirdesignandimplementation. First,thecomplexityofthetrialdesignarisesfromtheneedtointegrate adaptivedesigns,multistageprotocols,andcrossovertreatmentstrate-gies,whilebalancingthevalidationofcompositeendpoints,surrogate endpoints,andsurvivalbenefits.Second,significantheterogeneityin patientpopulationsanddiseasepathogenesisandprogression,suchas variationsintumorsubtypes,staging,andmolecularprofiles,complicate thepredictionoftherapeuticresponses,necessitatingbiomarker-driven stratificationtoenhanceprecision.Ethically,researchersmustnavigate the“therapeuticdesert”facedbyterminallyillpatients,dynamically weighingdrugtoxicity,qualityoflife,andpotentialclinicalbenefits,with particularemphasisonensuringtransparentinformedconsentand rigorousriskdisclosure.TherecentlyupdatedCONSORT2025statement reinforces standardized reporting requirements for complex

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interventions,patientsubgroupanalyses,andadverseeventgradingwith theaimofensuringtransparentandreliableevidence.Consideringthe distinctfeaturesofoncologyresearch,acontext-specificinterpretationof theCONSORTguidelinesisimperative.Theseconsiderationsdemand thatresearchersadheretoevidence-basedframeworksanddevelop oncology-specificmethodologicalreflections,ultimatelyfosteringthe synergisticadvancementofethicalrigorandscientificvalidityinclinical research.UsingtheupdatedCONSORT2025statement,thisarticleout-linesthelimitationsandchallengesfacedinoncologyclinicaltrialsand exploresadaptivesolutions.

Mainchangesinconsolidatedstandardsofreportingtrials2025

CONSORT2025addedsevennewitems,revisedthreeitems,deleted one,andintegratedandrestructuredsomeitemsbasedonthe2010 version[Table1].

TheupdatedCONSORTchecklistanddetaileditemdescriptionsare availableonitsofficialwebsite().More detailedexamplesandinterproviding

convenienceforunderstandingthelistofitems.Detailedinterpretations here.BasedontheCONSORT2025,wediscussthechallengesand gestionsfromdifferentperspectives.

ofeachitemhavebeenpublishedinChinese[9]andwerenotelaborated considerationsinreportingclinicaltrialsinoncologyandprovidesug-

Applicabilityandchallengeswithconsolidatedstandardsof reportingtrials2025inoncologyresearch

TheCONSORTstatementisageneralguidelineforreportingRCTs andprovidesaminimumsetofrecommendationsforreportingtrial details.ManyCONSORTextensionshavebeendevelopedtoaddressthe methodologicalissuesofreportingdifferenttrialdesigns,data,andin-terventions,includingrecommendationsforadaptivedesigns,[13]early phasetrials[14]multi-armtrials,[15]n-of-1trials,[16]andpatient-reported outcomes,amongothers.[17]

WhentheCONSORT2025statementisappliedtoanoncology clinicaltrial,theupdateditemsenablereaderstogainaclearerunder-standingofthetrialprocessandimprovethestudy'stransparency.For

Table1
InterpretationofmainchangesinConsolidatedStandardsofReportingTrials2025comparedtothe2010version.

Changestype	CONSORT2025item	Interpretation	Oncology-specificexamples
Addition	4	Clarifythedatasharedtoenhancerepeatability, includingwhereandhowindividualde-identifieddata werecollected,definitionsofvariables,analysiscodes,	•Sharegenomicrawdata(e.g.,FASTQfiles)and variantcallingcodesforbiomarker-driventrials•Provideprotocolsforliquidbiopsy ctDNAanalysis
	5b	andmaterialsrelatedtotheintervention.Ifsharingis	•Disclosepharmafundingfortargetedtherapytrials•ReportinstitutionalpatentsonCAR-Ttechnologies•Involvecancersurvivorsindesigning PROsfor
	8	notplanned,itshouldbeclearlystatedandexplained.	immunotherapyQoLassessment
		Emphasizetheconflictofinteresstatementsofall authors.Enhancethecredibilityoftheresearch.	•Engagecaregiversindefiningtoxicityreporting thresholds
		Expectthatpatientsorthepubliccanformpartnerships withresearchers,involvingthestudydesignand conduct,andreport,interpret,anddisseminatethetrial.	
	12b	Thepatientsorthepublicarenotparticipantsinthe trial.Theycanbeindividualswhocurrentlyorhavehad certainhealthconditions,aswellastheirfamily members,caregivers,socialworkers,andother membersofthepublic.Theparticipationofpatientsand	•SpecifyCAR-Tcellmanufacturingfacilitycertifications(e.g.,GMPcompliance) •DetailradiologistexperiencethresholdsforRECIST assessments
	15	thepublicisconducivetoimprovingtheefficiency and acceptabilityofrecruitmentandfollow-up,selectingthe outcomesandtheirmeasurementsthatthetarget populationtrulyneeds,andimprovingcommunication amongparticipantsandthedisseminationofresults. Considertheinfluenceoftheprovidercondition softhe interventionsontheeffect.Notallstudiesneedtoreport	•DefineCRSgradingpertheASTCTcriteriainCAR-T trials •Reportneurotoxicityonsettimelinesinbispecific antibodytrials
	21b,21c	thequalificationcriteriaoftheinterventionprovider.	•SensitivityanalysisforOSdatawithhighdropout in metastatictrials
		Whentheoutcomeoftheinterventionishighly associatedwiththecharacteristicsoftheintervention provider,suchashumanresources,technicallevels,and management,thetherapeuticeffectwillbeaffected. Adverseeventscanincludepre-consideredandnon-pre-	•Multipleimputationformissingbiomarkerdatainneoadjuvantstudies

(*continuedonnextpage*)

Table1(*continued*)

Changestype	CONSORT2025item	Interpretation	Oncology-specificexamples
	24	comparisonwiththeobserveddatacharacteristics, whetherinterpolationwasadopted,thespecificinterpolationmethod,andmoreimportantly,theresults ofsensitivityanalysis,andtheimpactofmissingdataon theinterpretationoftheresults.Atleastasummaryof thesensitivityanalysisofthemissingdatashouldbe reported.Thedetailedresultsshouldbegiveninthe supplementarymaterials.	•Reportdeviationsinradiotherapydosimetry•Documentconcomitantuseofantifungals during AMLchemotherapy
Revision	3	Theinterventiontakenduringthetrialmightdifferslightlyfromthatatthedesignstage,whichmaybedue toexecutiveability,compliance,interventiondiversity, andcomplexity,etc.Concomitanttreatmentsoutside thetrialinterventionmightintroducepost-randomizationbias,especiallywhentheconcomitant treatmentsaresignificantlyassociatedwiththeprimary outcome,havingagreaterimpactonthereliabilityof theresults.Itisnecessarytofullyreportthe	•Pre-specifysubgroupanalysesforPD-L1expression levels •LockSAPbeforetheinterimanalysisofPFSendpoints
	10	considerationsatthedesignstage,thedisparityfrom	•Flagpost-hocanalysesofTMBpredictivevalue•Justifyaddedendpointslike MRDnegativityin protocolamendments
	26	thedesignstage,thetypeandfrequencyofthe accompanyingtreatment,and,moreimportantly,the differencesbetweengroups.	•Reportattritionratesat24-monthOSfollow-up•ProvidereasonsformissingRECISTassessments(e.g., COVID-19scandelays)
Deletion	30	Emphasizeaccesstopre-plannedSAP.Beforethe participantallocation,datacollection,ordataset for analysis,anychangeinthetrialprotocolandSAPdueto unexpectedchallengesortheemergenceofnew evidenceshouldbereviewedandsigned.	•Discussapplicabilityofbaskettrialresultstorare subtypes(e.g.,NTRKfusions)
Integrationandrestructuring	Safety/adverseevents 7,15,21a,23a,27 14,26	Reportsonun-pre-specifiedoutcomesoranalyses.The changeoftheplanwouldaffectthetrialdesignand conduct.	•UnifiedreportingofirAEsacrossCTCAEcategories
	24	Thenumberofparticipantswithavailabledataat each timepointforeachprimaryandsecondaryoutcome	
	Openscience2–5	was emphasized.Ifpossible,thereasonsforthe unavailabilityofdatashouldbereported,suchas lossto follow-uporcensoring.Therelativeeffects(hazard ratiooroddsratio)andabsoluteeffects(riskdifference)	
	All	forthebinaryprimaryoutcomeshouldbereport	

AEs:Adverseevents;AML:Acutemyeloidleukemia;ASTCT:AmericanSocietyforTransplantationandCellularTherapy;CAR-T:Chimericantigenreceptor-therapy; CONSORT:ConsolidatedStandardsofReportingTrials;COVID-19:Coronavirusdisease2019;CRS:Cytokinereleasesyndrome;CTCAE:Commonterminologycriteria foradverseevents;ctDNA:CirculatingtumorDNA;FASTQ:Themostcommonlyusedfileformatforstoringnucleotidesequenceinformationorproteinsequence information;GMP:Goodmanufacturingpractice;irAEs:Immune-relatedAEs;ITT:Intention-to-treat;MRD:Measurableresidualdisease;NTRK:NeuroTrophinReceptorKinase;OS:Overallsurvival;PD-L1:Programmeddeath-ligand1;PFS:Progression-freesurvival;PROs:Patient-reportedoutcomes;QoL:Qualityoflife;RCTs: Randomizedcontrolledtrials;RECIST:Responseevaluationcriteriainsolidtumors;SAP:Statisticalanalysisplan;SPIRIT:StandardProtocolItems:Recommendations forInterventionalTrials;TMB:Tumormutationalburden.

instance,thedatasharinginitem4canpromotedatatransparency,such asliquidbiopsyandgenesequencinginvolvedintumor-relatedtrials. Patientsandpublicparticipationinitem8willenhancetheconsideration ofpatients'feedbackonindividualizedtherapy(suchaschimeric antigenreceptorT-celltherapy[CAR-T]),toreflectthedevelopment trendof“patient-oriented.”Theharmstatementinitem15requires systematicreportingofuniquesafetyeventsduringimmunotherapy (suchascytokinestorms).However,significantreportinggapspersistin

non-oncology-specificextensionsinfullyaddressingthereportingneeds ofthisfield.Inthesectionbelow,wesummarizethefollowingsixas-pectsanddiscusstheissuesthatneedtobeconsideredandreportedin oncologytrials.

Selectionofbiomarkers

Biomarker-drivendesignshavebeenappliedinoncologicalclinical trials,includingscreeningandenrichmentstrategies.Thereisalackof reportingstandardsfortargetedtherapyinpatientsundergoinggene mutationscreening.Keyinformation,includingbiologicalandclinical rationales,biomarkerprevalence,safetydata,biomarkervalidation methods,overalltrialdesign,andsubgroupefficacycharacterization, shouldbereportedindetail.Currently,CONSORTanditscorresponding extensionsdonotincluderelevantrequirementsregardingtheselection ofbiomarkers,thedetectionprocess,ortheinterpretationofresults.To addressthisgap,weproposethefollowingminimumreportingstan-dardsforbiomarker-drivenoncologytrials[SupplementaryTable1].

Complexityofcombinationtherapy

Combinationorintegratedtherapyiscurrentlyapopulartopicanda futuredevelopmenttrend.Theantitumormechanismsofdifferent treatmentstrategiesvary;however,theremayalsobesynergisticor antagonisticeffectsamonginterventionmechanisms.Inclinicaltrials involvingmultipleinterventionmechanisms,theinteractionsbetween differentinterventionsshouldbeanalyzedandreportedextensivelyto ensureclearercausalinferences.Ancillaryanalysesinitem28seemedto includeaninteractionanalysis.Additionally,thetheoreticalbasisof specificcombinedorintegratedtherapiesshouldbereportedindetail.

Implementationandsafetyofcomplexinterventions

Recently,immunotherapywithcomplexprocesseshasconflicted withstandardizedinterventionsinclassicRCTs.Forexample,CAR-T treatmentinvolvescellextraction,preparation,andreinfusion.

Genetherapymayrequireinterventionsatmultiplestages,suchas viralvectordeliveryandgeneediting.Operationandqualitycontrolat eachstepmayaffecttheoutcomes.Theimplementationofsuchdetailed interventionsandqualitycontrolisparticularlyimportantforthe interpretationoftrialresults.Formorecomplexinterventions,CON-SORTalreadyhascorrespondingprinciples,namely,thedetailedelab-orationoftheinterventionimplementationinitem24.Individualized interventionimplementationandcorrespondingqualitycontrolmea-suresshouldbedescribedindetail.

Selectionofoutcomeendpointsforhighlyheterogeneoustumors

Item14clearlydistinguishesbetweenprimaryandsecondaryout-comesandexplainsclinicalrationality.However,themainoutcomesof efficacyevaluationsinoncologyclinicaltrialsareheterogeneous.For example,somesolidtumorsuseprogression-freesurvival(PFS)asthe primaryendpoint,whereasforhematologicaltumors,moreattention maybepaidtothecompleteresponserate(CR).Domesticandforeign regulatoryauthoritieshaverecentlyissuedcorrespondingguiding principlesfortheuseofmeasurableresidualdisease(MRD)tobeusedas asurrogateforspecifictumors.Inthisregard,onthebasisofthe followingrelevantprinciples,thebasisforoutcomeselectionshouldalso beclearlystatedinthereport.

Standardizedapplicationofemergingtechnologies

Manyemergingtechnologies,includingliquidbiopsy,artificial intelligence-assisteddiagnosis,andmethodssuchasdeeplearningal-gorithmsorrecurrenceandcompositeoutcomeanalysis,haveemerged; however,theCONSORT2025statementhaslimitedcoverage.Usingthe

columntopromotethepopularizationandapplicationofreporting standardization,andthedevelopmentofreportingguidelinesspecified foroncologicalclinicaltrials.Forajournalarticleofseveralthousand words,consideringthelengthofthearticle,itisunrealistictoincludeall itemsrelatedtothestatisticalmethodsinthemaintext.However, journalscanrequestrelevantlinks,sothatinterestholderscanobtain detailedstatisticalanalysismethods,plans,andcodes.Thisalsoapplies tointerventions.Acompletereportshouldnotbeconfinedtoafew pagesofajournal.

Finally,toensurewidespreadcomprehensionandeffectiveutilization ofthereportingstandardizationtools,structuredtrainingprograms onCONSORT2025shouldbeimplementedforthediversestakeholders involvedinoncologytrials.TheCONSORTGroup,potentiallyin collaborationwithmajoroncologysocietiesorjournals,shoulddevelop standardizedtrainingmodules(e.g.,onlinecourses,webinars,down-loadableresources)coveringthecoreprinciplesandspecificextensions relevanttooncology.Institutions,contractresearchorganizations (CROs),andregionalconsortiashouldthenconducttargetedworkshops andhands-onsessionstailoredtospecificroles(e.g.,investigators, researchcoordinators,datamanagers,statisticians,andmedicalwriters) tofacilitatepracticalapplications.

academicsocietiesandassociations,expertsfromvariousfieldsare requiredtodiscussoradoptcorrespondingscientificmethods,conduct research,formspecificsupplementaryitemsorextendedversions,and promotetheestablishmentofaclinicalresearchdatasharingplatformin oncologicalresearch,facilitatingtransparentresearchandopenscien-tificpractice.Throughthejointeffortsofresearchers,journals,and guidelinemakers,abalancebetweenstandardizationandinnovationin high-qualityoncologicalclinicaltrialreportscanbeachieved.

Authorscontribution

YaguangPeng:Informationcollectionanddatacuration,analysis, andwritingoriginaldraft;PengLyu:Projectadministrationandwriting-review&editing;XiaoxiaPeng:Conceptualizationandmethodology, andwriting-review&editing.Alltheauthorshavereadandapproved thefinalpaper.

Ethicsstatement

None.

Futuredirectionandappeal

ItisimportanttoemphasizeagainthattheCONSORTstatement providesabasicframeworkforreportingclinicaltrials,constitutingthe minimumrequirementtoensureclarity,transparency,andultimately, thereflectionoftheauthenticityandreliabilityofthetrialdesign, implementation,andanalysis.High-qualityreportingisafundamental prerequisiteforreproducibility.[19]Giventhepersistentandcritical reportinggapsidentifiedinoncologytrials,coupledwiththefield's uniquemethodologicalandpracticalcomplexities(e.g.,biomarker stratification,combinationtherapieswithpotentialinteractions,com-plexinterventionssuchascellulartherapies,diverseendpointselection, andemergingtechnologies),thedevelopmentofadedicatedCONSORT extensionforoncologicalclinicaltrials(CONSORT-Oncology)isnot onlywarrantedbuturgentlyneeded.Suchanextensioncouldprovide tailoredguidanceforaddressingthespecificchallengesoutlinedinthis study.Forinstance,inbiomarker-driventrials,detailsshouldbepro-videdregardingthebasisforbiomarkerselection(fromaclinicaland/or biologicalrationale),validationmethods(suchassequencingplatform certification),andresultinterpretation(suchasfalse-positiveratecon-trols).Regardingtheadoptionofcombinedorintegratedtherapies,in-teractionsamonginterventionsshouldhavebeenconsideredand reflectedinstatisticalanalysismethodsandresults.Furthermore,when applyingemergingtechnologies,itisnecessarytodisclosethetechnical detailsofdetection(suchasthesensitivityofcirculatingtumorDNA [ctDNA]detection,sequencingdepth),training,andverificationpro-cessesoftheartificialintelligence(AI)modelprediction.

Extendingequityiscrucialforethicsandsocialequity.Consider includingextendeditemssimilarto“EconomicBurdenandResource Allocation,”whichnecessitateananalysisofdifferencesintreatment costs,insurancecoverage,andregionalaccessibility.Forinstance,dis-paritiesinaccesstooncologyclinicaltrialsarewell-documentedand significantlyimpactequitablecare.Studieshaveshownthatfactorssuch associoeconomicstatus,geographiclocation(e.g.,distancefrommajor academiccenters),race/ethnicity,andinsurancetypecreatesubstantial barrierstotrialparticipation.[20,21]Integratingspecificmetricsrelatedto thesedimensions,suchasthereportingofpatientdemographics (includingzoneimprovementplancodeorregion),insurancestatus,and trialenrollmentratesstratifiedbyrelevantfactors,wouldstrengthen reportingstandardsanddirectlyaddresstheequityconcernsraisedby high-costtherapiessuchasCAR-T.

Finally,throughinterdisciplinarycollaborationwithexpertsinbio-informaticsandethics,theframeworkoftechnicalstandardizationand ethicalreportingcanbeimproved.Throughthecoordinationof

Dataavailabilitystatement

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